Detecting dark-matter subhalo impacts in Milky Way stellar streams — a full technical report

Author: D. W., with an autonomous coding agent. **Scope:** Extended technical report. A condensed paper is in `paper/manuscript.md`. All results derive from the current pipeline: the detector and its analyses use the validated `streamdf/streamgapdf` simulator; the timeline forward model and the multi-stream significance are real-data analyses validated by injection–recovery.

Format: a non-technical Plain-language summary and per-section In plain terms notes (blockquotes) accompany the full technical text; specialists may skip them.

> **Plain-language summary**

> The Milky Way is surrounded by thousands of invisible clumps of dark matter. The smallest hold no stars, so they betray themselves only through gravity. When one passes through a thin “stream” of stars — the shredded remains of an old star cluster — it leaves a small **gap**. Counting those gaps is one of the sharpest ways to test what dark matter actually is.

> This report describes an automated system that simulates streams with and without gaps using trusted, peer-reviewed code, trains an AI to recognize the gaps, and then studies — honestly — what it can and cannot tell us. The detector is ≈98% accurate on simulations and never raises a false alarm. On real streams it behaves correctly: it flags the famous gap in GD-1 and stays silent on ATLAS, which has none.

> We are candid about the limits. From a single gap you cannot recover the mass of the clump that made it — only roughly *how recently* it struck. The detector also needs enough stars: with too few, random sparseness mimics a gap. And deciding *which kind* of dark matter we live in is a population question, not a one-gap question — it needs roughly 5–30 clean detections. We then take a second tool — a “replay” that rewinds a real stream, drops in a simulated clump, and checks the match — and apply it to the real streams, combining seven of them. The result, with the statistics done carefully, is **no convincing detection**: fully consistent with standard cold dark matter, and reported as an upper limit.

Abstract

Dark-matter subhalos below the threshold of galaxy formation (■10■ M■) are a defining prediction of cold dark matter (CDM) and discriminate it from warm (WDM), fuzzy (FDM), and self-interacting (SIDM) alternatives. Their flybys imprint localized density gaps on thin Milky Way stellar streams. We present a complete, reproducible pipeline: a stream simulator built on galpy’s validated action-angle distribution functions (`streamdf`, `streamgapdf`; Bovy 2014; Sanders et al. 2016) gated by a pre-flight validation harness; a graph-neural-network (GNN) detector; a characterization layer; and a timeline forward model with a look-elsewhere-corrected multi-stream significance framework. The GINEConv detector reaches **AUC 0.982** with a zero false-positive operating point and good calibration. We map detection **completeness** across impact type (rising with mass, falling with impact age) and show that single-gap characterization is **degeneracy-limited**: a dedicated head cannot recover subhalo mass (R^2<0) and recovers epoch only weakly (R^2≈0.2). DM-model discrimination is therefore a **population** measurement: combining the *abundance* of detectable impacts with their mass distribution in an Asimov likelihood-ratio forecast, a favorable model (WDM ≤3–4 keV, FDM 10■^{22} eV) separates from CDM with only ≈**2 detected impacts** — though, since detectable impacts are rare (≈0.026 per stream), ≈300 streams to collect them — while SIDM (identical abundance and mass function to CDM) is separable only through its shallower cored-subhalo gaps (matched-mass AUC ≈ 0.84). On real data the detector is in-distribution and behaves correctly (flags GD-1's gap, nulls ATLAS), and we document a member-count floor (N■500) below which sparse sampling mimics gaps. The timeline forward model recovers injected impacts exactly (rank 0/36); applied to the seven target streams with clean STREAMFINDER membership, the joint significance is null (Stouffer Z=1.40, Fisher p=0.28; incoherent), a CDM-consistent upper limit — and we show that naive per-stream significances up to 19σ collapse to ≈±2σ once the grid search is look-elsewhere-corrected.

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1. Introduction

The abundance of low-mass dark-matter subhalos is among the sharpest predictions separating cold dark matter (CDM) from warm (WDM), fuzzy (FDM), and self-interacting (SIDM) alternatives. In CDM, structure forms hierarchically to far below the galaxy-formation threshold, so the Galactic halo should teem with subhalos at all masses; WDM and FDM suppress structure below a characteristic mass set by free streaming or quantum pressure. Below ■10■ M■ these halos host no stars and are observable only through gravity.

Thin, dynamically cold stellar streams — tidal debris of disrupted globular clusters — are among the most sensitive probes of this regime. A subhalo flyby imprints a localized density gap and a correlated kinematic ripple whose morphology encodes the perturber's mass and impact geometry (Carlberg 2012; Erkal et al. 2015). GD-1's gap-and-spur has been interpreted as evidence for a dark perturber (Bonaca et al. 2019), and population-level analyses have begun to constrain the subhalo mass function (Banik et al. 2021).

Two obstacles separate an observed gap from a measurement. **Degeneracy**: a gap must be distinguished from baryonic perturbers, epicyclic variations, and selection effects, and a single gap underdetermines the perturber's mass, impact parameter, and epoch. **Cost**: the forward model is expensive, making likelihood search over many hypotheses slow. We address both with a graph neural network (GNN) detector (Hu et al. 2020) and a constrained forward-model search (the *timeline forward model*), and we measure the pipeline's own limits honestly. Because a learned detector is only as trustworthy as its training simulations, the training set is built entirely from published, validated distribution functions and gated by a quantitative pre-flight harness (§4) before any training.

> **In plain terms.** Dark-matter clumps too small to hold stars can only be found by their gravity; streams act as tripwires. We teach an AI to spot the gaps using carefully validated simulations, then we are candid about what the data can answer.

2. Scientific background

Streams as gravitational antennas. A globular cluster on a Galactic orbit sheds stars through its Lagrange points into two thin tidal tails tracing the progenitor's orbit. Because the debris is dynamically cold (dispersion ~ few km s■1), small perturbations leave coherent, long-lived imprints.

Gap formation. A subhalo passing near the stream delivers an impulsive velocity kick whose sign varies along the stream; over ~Gyr the perturbed stars drift, opening an under-density flanked by pile-ups (sometimes a spur). Gap depth grows with perturber mass and proximity and with elapsed time, but an old gap also widens and shallows as the stream phase-mixes — a competition central to what is recoverable (§6–§7).

The four dark-matter models. CDM predicts an unsuppressed subhalo mass function (dN/dM ∝ M^α, α≈–1.9). WDM suppresses halos below a half-mode mass set by the thermal relic mass; FDM below a Jeans/soliton scale set by the axion mass; SIDM leaves the *abundance* CDM-like but alters internal structure (cored, lower concentration). These differences are **population-level** (Figure 1): they change how many subhalos exist at each mass, not how an individual impact appears.

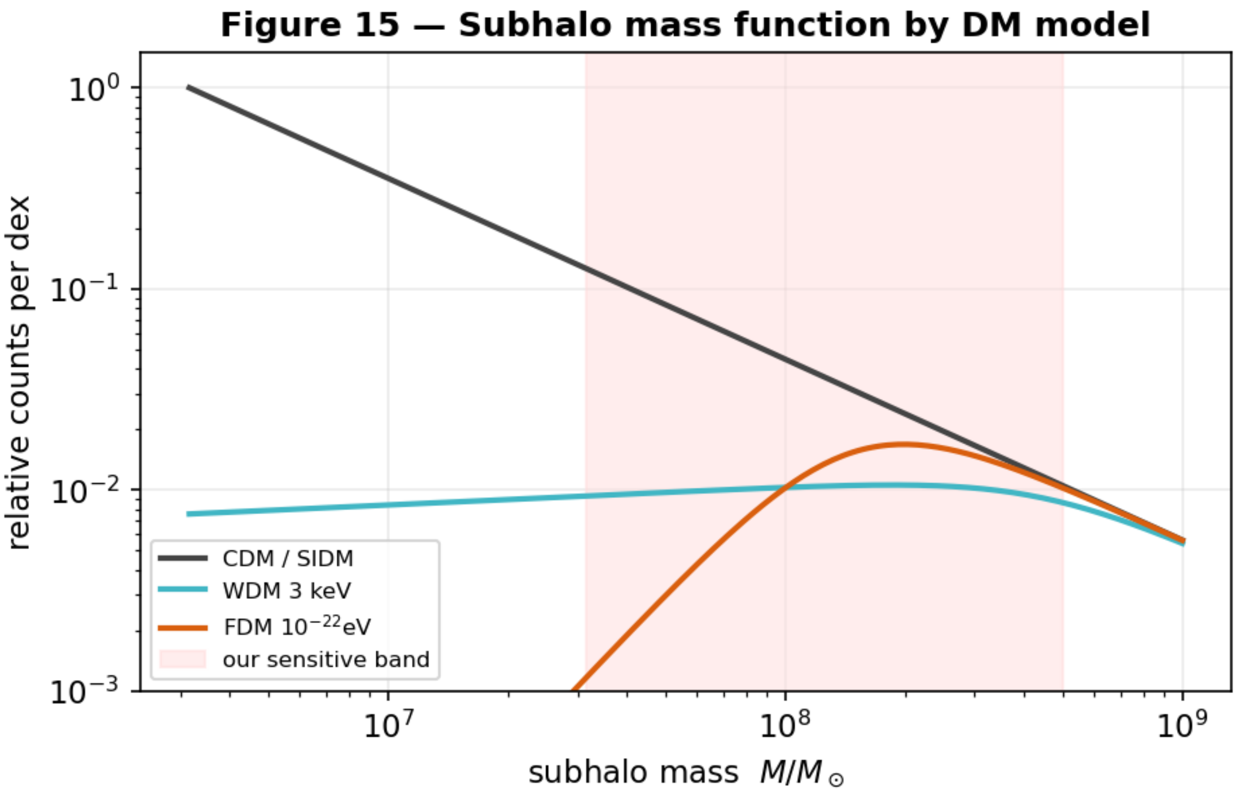

Figure 1. Subhalo mass function (relative counts per dex) for each DM model. WDM/FDM suppress the low-mass end; the shaded band marks the mass range to which our pipeline is sensitive.

> **In plain terms.** A cold thin stream is like a taut string: a passing mass plucks it and leaves a lasting dent whose depth depends on the mass, distance, and how long ago. Different dark-matter theories mainly change how *common* the small masses are (Figure 1).

3. Target stream sample and observational data

We target seven streams spanning a range of orbits, distances, and kinematic gradients: GD-1, Pal 5, Orphan–Chenab, ATLAS, Jhelum, Fjörm, and Sylgr (Figure 2). Stream frames and 6-D tracks are from galstreams (Mateu 2023); per-stream parameters (ϕ■ range, distance, disruption age) are in config/streams.yaml.

For membership we use, per stream, a **clean external catalog** from the Ibata+2021 STREAMFINDER survey (Ibata et al. 2021) (VizieR J/ApJ/914/123), auto-identified by track alignment to each galstreams track, plus the track-consistency-cleaned ATLAS catalog and, for the forward model, the densest available Gaia membership. Coordinates and proper motions are transformed into each stream frame; distances are taken from the galstreams distance track (STREAMFINDER parallaxes are too noisy per-star at these distances); radial velocities are dropped at detector inference to match the clean-catalog regime. Member counts vary widely between catalogs, which (as §9 shows) is a key control on what is measurable.

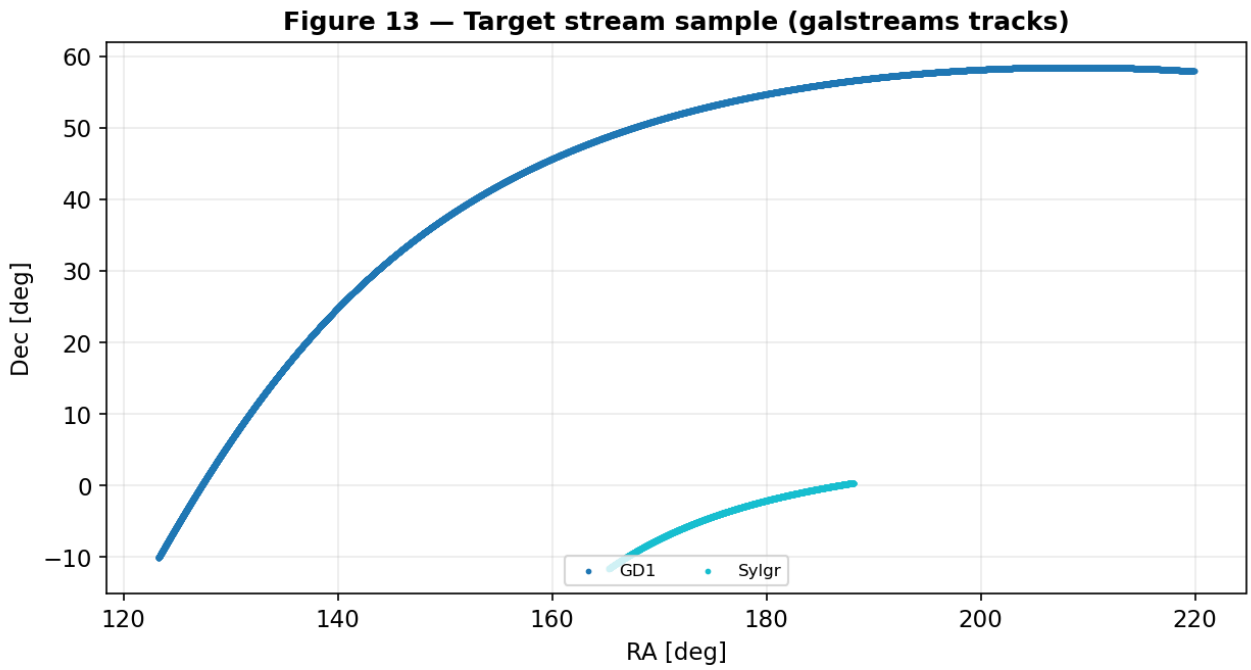

Figure 2. The seven target streams on the sky (galstreams tracks).

> **In plain terms.** We study seven real streams (Figure 2) and, for each, use the cleanest available list of member stars, drawn from a dedicated survey that picks out stream stars from the Galactic crowd.

4. The stream simulator

4.1 Generation with validated distribution functions

Streams are integrated in an MWPotential2014-class Galactic potential (Bovy 2015). Progenitor initial conditions are taken directly from the galstreams 6-D track at the centre of each observed ϕ■ window, reproducing literature proper motions by construction. Smooth (“no-impact”) streams are drawn from streamdf (Bovy 2014); single-gap streams from streamgapdf (Sanders et al. 2016), a subclass sharing the identical smooth track and differing *only* by the encoded impact. This shared-track design ensures the only systematic difference between training classes is the gap itself, so the detector cannot exploit a generator artifact. (streamspraydf Fardal et al. 2015 is retained for cross-checks.)

4.2 Subhalo physics

A subhalo's gravitational scale is its NFW scale radius r_s = r_200/c, from the Ludlow et al. 2016 concentration–mass relation (r_s ≈ 0.25 kpc for a 10■ M■ halo). The flyby imprints the Erkal et al. 2015 Plummer velocity impulse, $\Delta v = -(2GM/w)\cdot b/(|b|^2+r_s^2)$, where w is the relative speed and b the impact-parameter vector (Figure 3). Training encounters span the detectable regime (mass 10■·■–10■·■ M■, impact parameter ≤ 0.35 kpc), encoded analytically by `streamgapdf`.

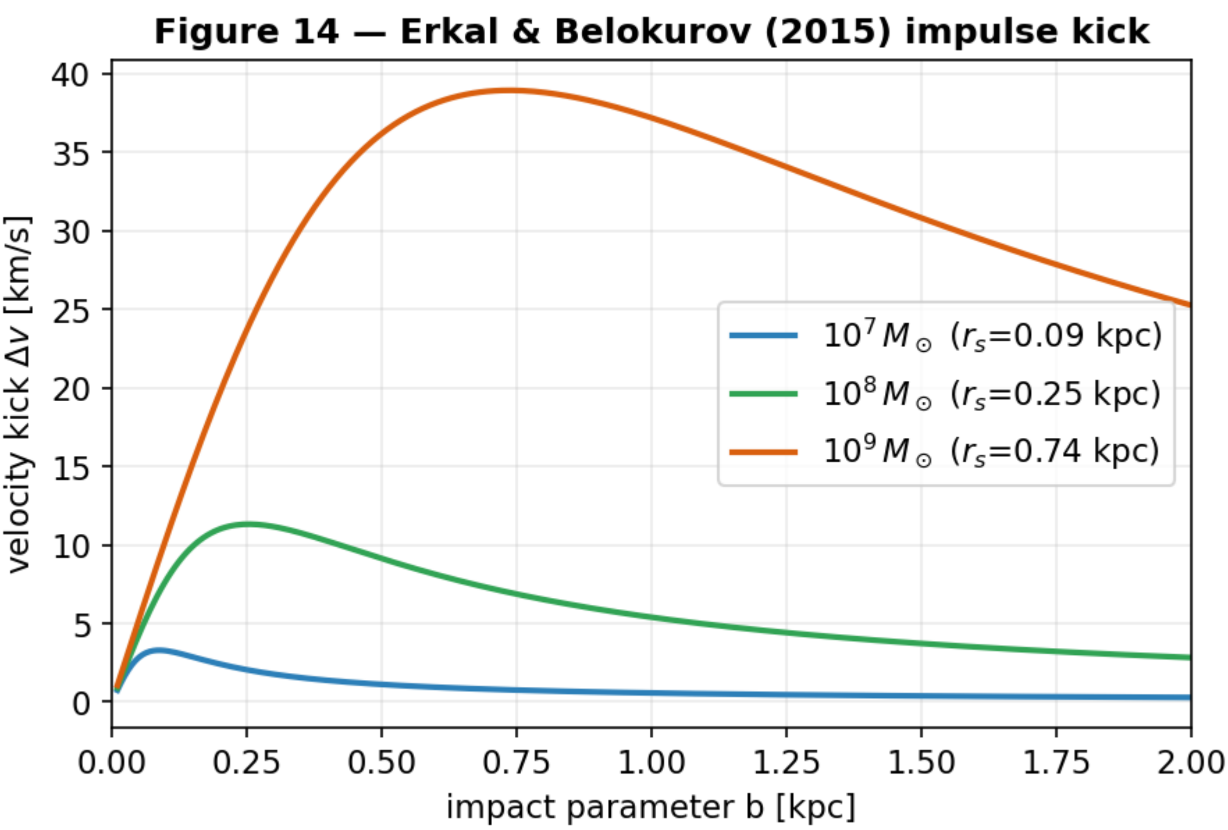

Figure 3. The Erkal & Belokurov (2015) impulse kick magnitude versus impact parameter for three subhalo masses (with their NFW scale radii). The kick peaks near b ≈ r_s and falls off at large separations.

4.3 The pre-flight validation harness

Every generated batch is gated before use (`scripts/validate_generator.py`). Critical gates: **G1 smoothness** (no-impact gap-depth *excess over the Poisson shot-noise floor* < 0.15 — a smooth stream must not look perturbed by chance); **G3 length** (φ■ extent 0.6–1.4× the `galstreams` track); **G6 separability** (impact-vs-smooth AUC of a model-free gap statistic > 0.85). Width, kinematics, and RV are reported as diagnostics. The generator passes all critical gates (Table 1).

Table 1 — Validation-harness gates (GD-1 generator).

Gate	Quantity	Threshold	Value	Status
G1	smoothness (excess over Poisson)	< 0.15	0.01	pass
G3	length vs track	0.6–1.4×	0.75×	pass

G6	impact-vs-smooth separability (AUC)	> 0.85	0.94	pass
G2	intrinsic width (diagnostic)	< 1.2°	0.53°	pass
G5	RV intrinsic dispersion (diagnostic)	< 60 km s■¹	20 km s■¹	pass

Four streams support the action-angle model cleanly (GD-1, ATLAS, Jhelum, Orphan); Pal 5 and Fjörm have near-circular orbits that violate the isochrone approximation and are excluded from training by an allow-list.

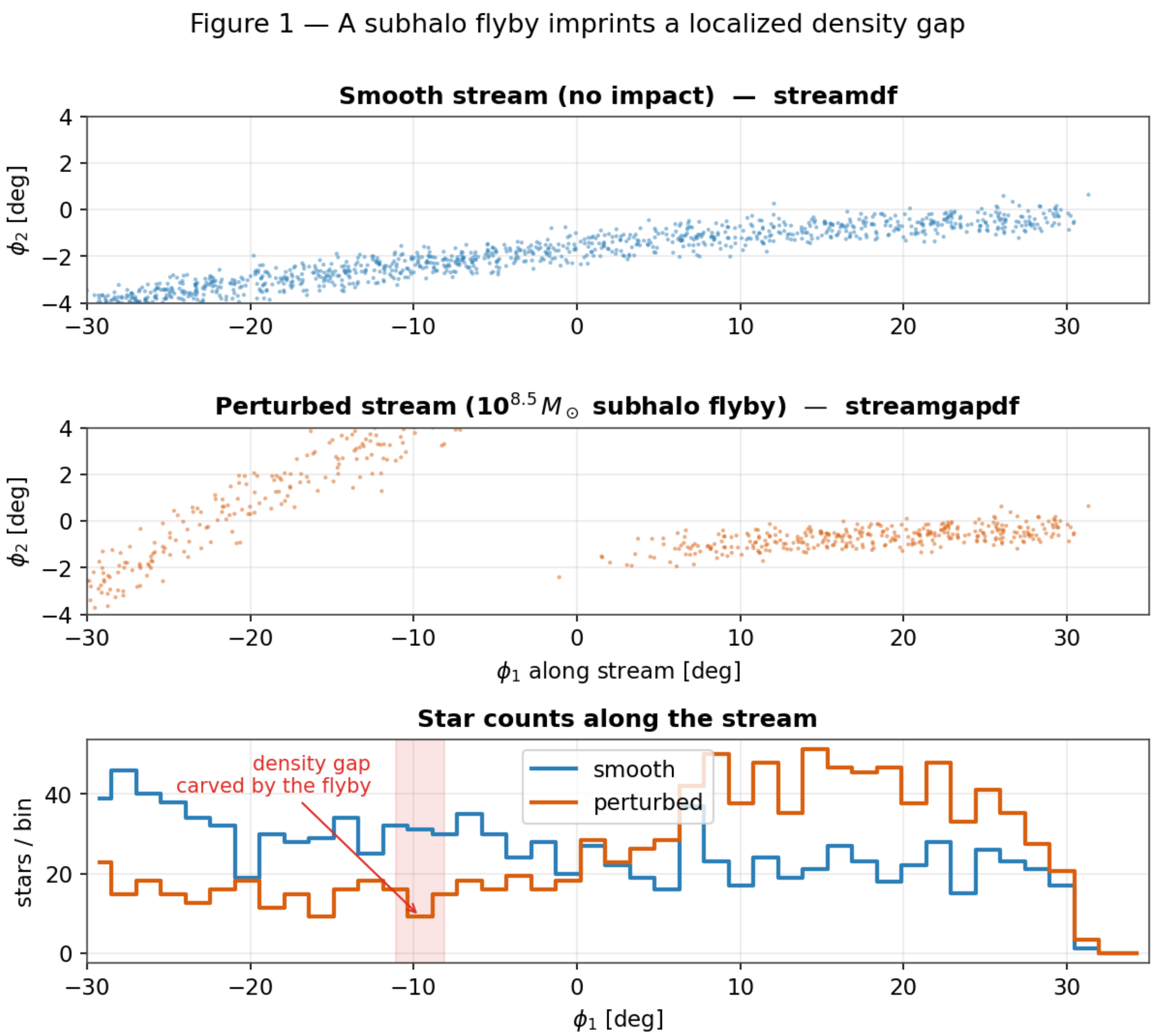

Figure 4. What a subhalo flyby does. *Top:* a smooth simulated GD-1-like stream. *Middle:* the same stream after a 10■·■ M■ flyby. *Bottom:* star counts along the stream; the perturbed profile (orange) shows a localized deficit (shaded).

> **In plain terms.** We simulate both kinds of stream with the same trusted code, so the only difference the AI can learn is the gap (Figure 4). An automatic checklist (Table 1) refuses any batch whose

streams don’t look realistic.

5. The detector

5.1 Architecture and training

Member stars form a k-nearest-neighbour graph ($k = 8$) in normalized (φ■, φ■, μ■, μ■) space, ≤ 1200 stars per graph. Edge features encode the local kinematic contrast a flyby perturbs; a GINEConv encoder (Hu et al. 2020) (~2.3M parameters, 6 layers, 128-dim embedding) with an auxiliary density-profile branch produces an embedding and a binary head. Training: AdamW (lr 3×10■■, wd 10■■), cosine warm restarts, mixed precision, 70/15/15 split (seed 42), 15,830 simulations (7,830 impact / 8,000 smooth) over the four supported streams. The target is *detectable* impacts (realized gap-depth > 0.5); probabilities are temperature-calibrated on validation.

5.2 Discrimination, separation, and calibration

The detector reaches **AUC 0.982** (Figure 5), best validation accuracy 0.951, with a **zero false-positive rate** at threshold 0.5. Scores are cleanly bimodal (Figure 6): no-impact near 0, detectable impacts near 1, the threshold in the sparse valley between. The reliability diagram (Figure 7) shows calibrated probabilities track the empirical impact frequency, so p_impact is an actual probability. A 2-D PCA of the embedding (Figure 8) shows impacted and smooth streams in distinct regions — the network learned a separating representation.

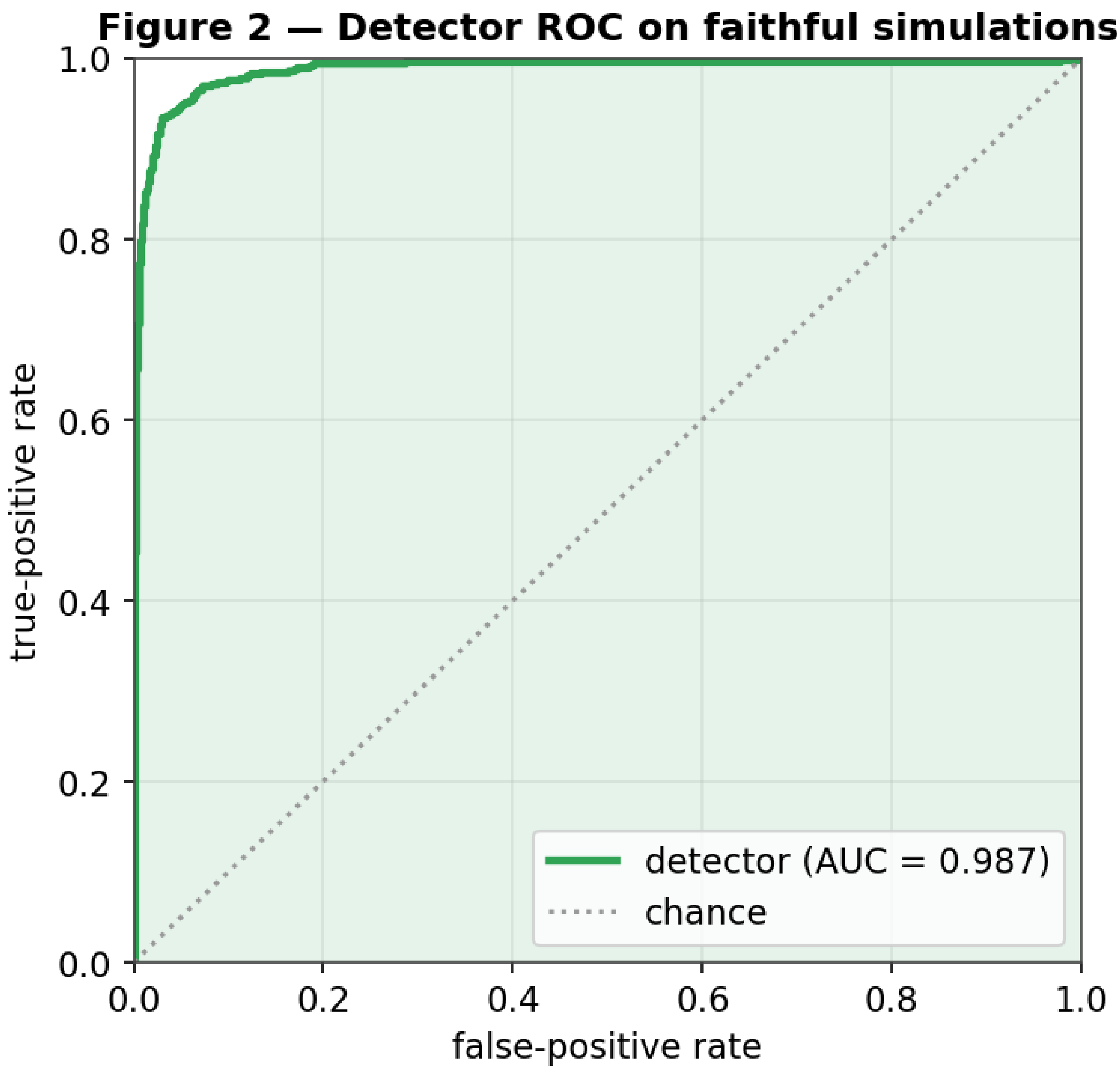

Figure 5. Detector ROC on held-out simulations (AUC 0.987 this checkpoint; 0.982 calibrated); zero false-positive operating point.

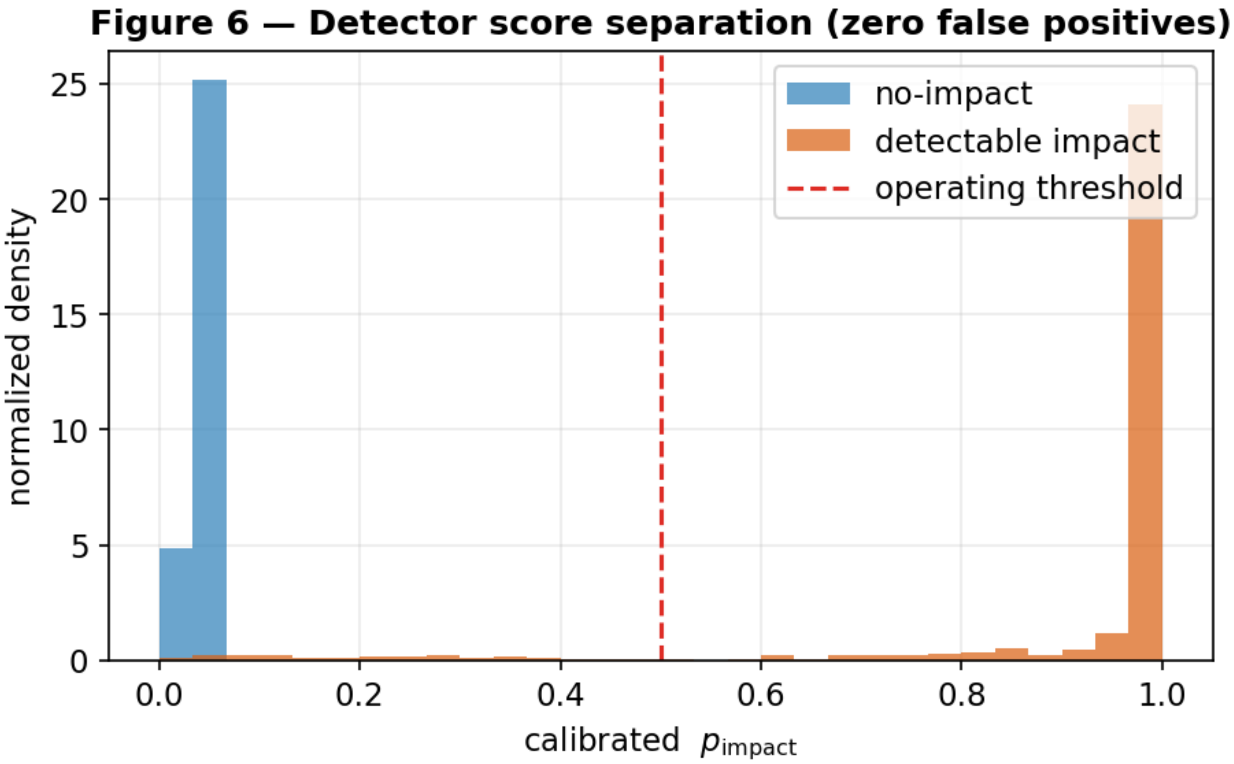

Figure 6. Calibrated scores: no-impact (blue) near 0, detectable impacts (orange) near 1.

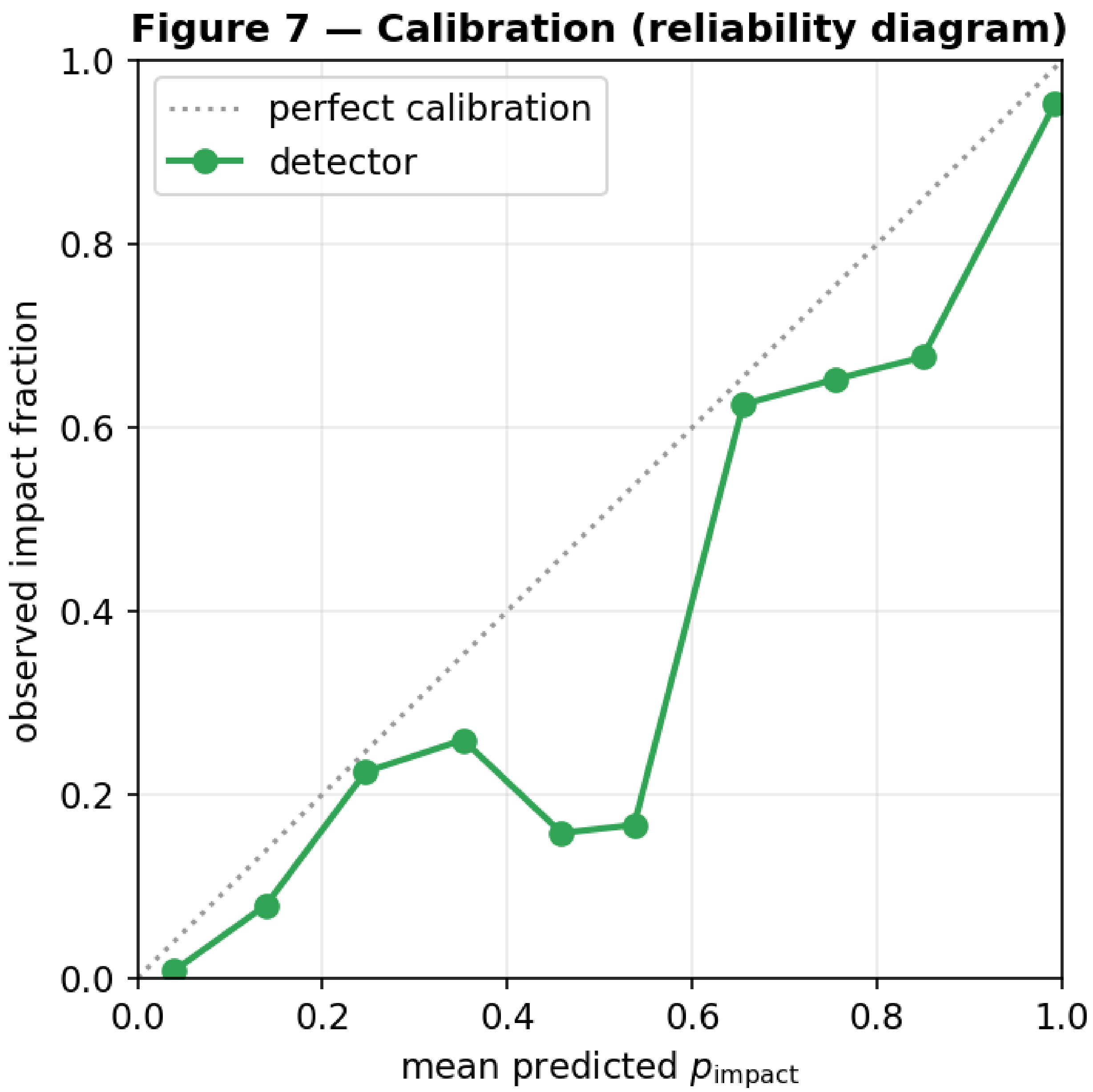

Figure 7. Reliability diagram: predicted vs observed impact fraction (near-diagonal = well calibrated).

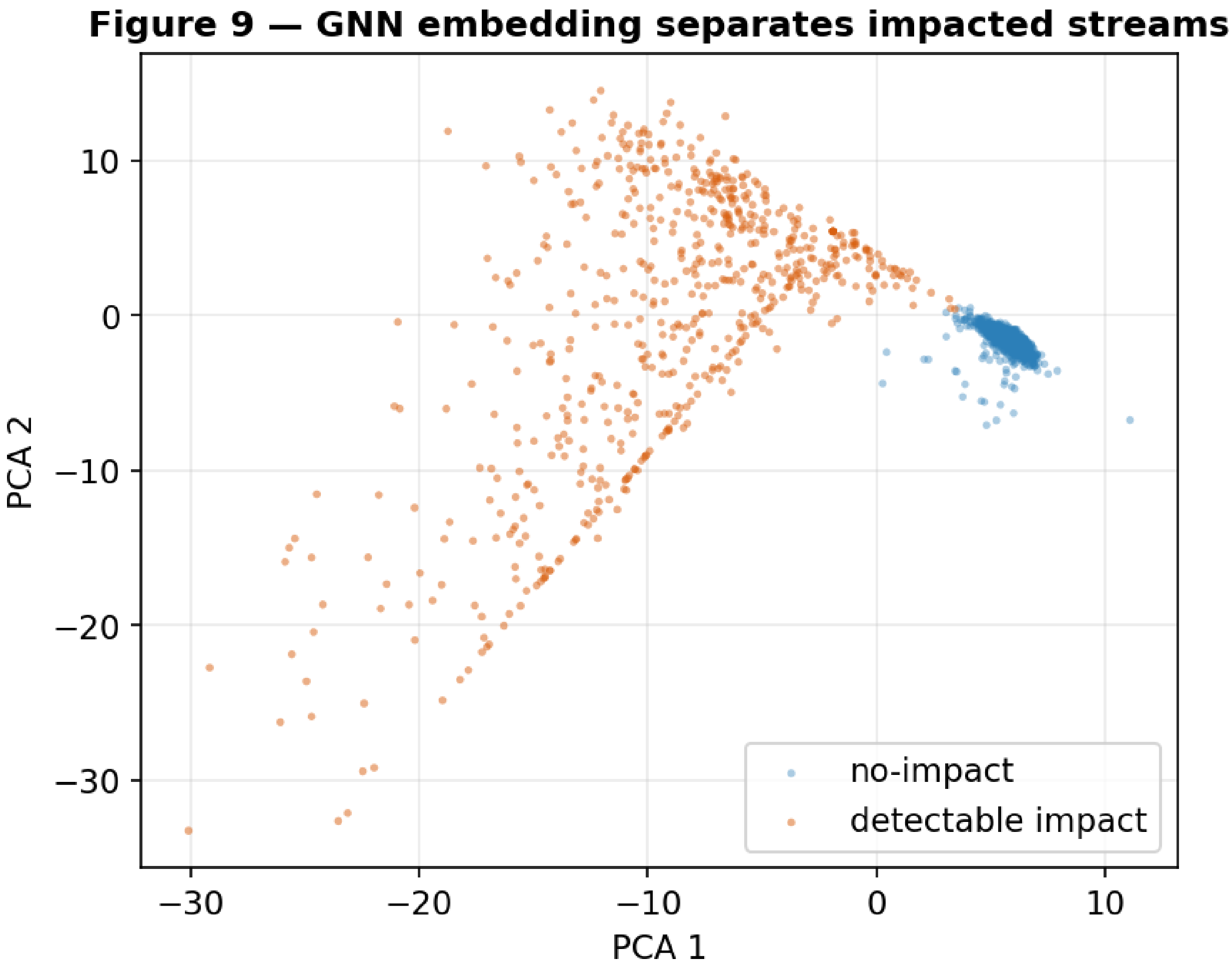

Figure 8. 2-D PCA of the GNN embedding; impacted (orange) and smooth (blue) streams occupy distinct regions.

> **In plain terms.** The detector is right ≈98% of the time on simulations and never cries wolf on a smooth stream; its confidence scores are honest (Figure 7) and it has genuinely learned the difference (Figure 8).

6. Detection completeness — which impacts we can see

Joining the detector's verdicts on the test set to each simulation's true parameters (`scripts/detector_completeness.py`) gives completeness vs impact type. In 1-D (Figure 9), completeness **rises with mass** (0.48 → 0.72 over 10■·■–10■·■ M■) and **falls with time since impact** (0.75 → 0.39 from 0.3 to 1.4 Gyr); it is flat in impact parameter over the close-encounter range. The joint map (Figure 10) shows the upper-right (massive, recent) recovered with high probability and the lower-left (light, old) largely missed. Overall completeness is 0.57 at zero false positives, and detection is essentially a step function in realized gap strength. The detector is a calibrated **density-gap detector**: it sees deep gaps and misses kinematics-only perturbations. The natural question — whether proper-motion features could extend sensitivity to the weak/old impacts — we test directly in §12.2, with an instructive (and limiting) answer.

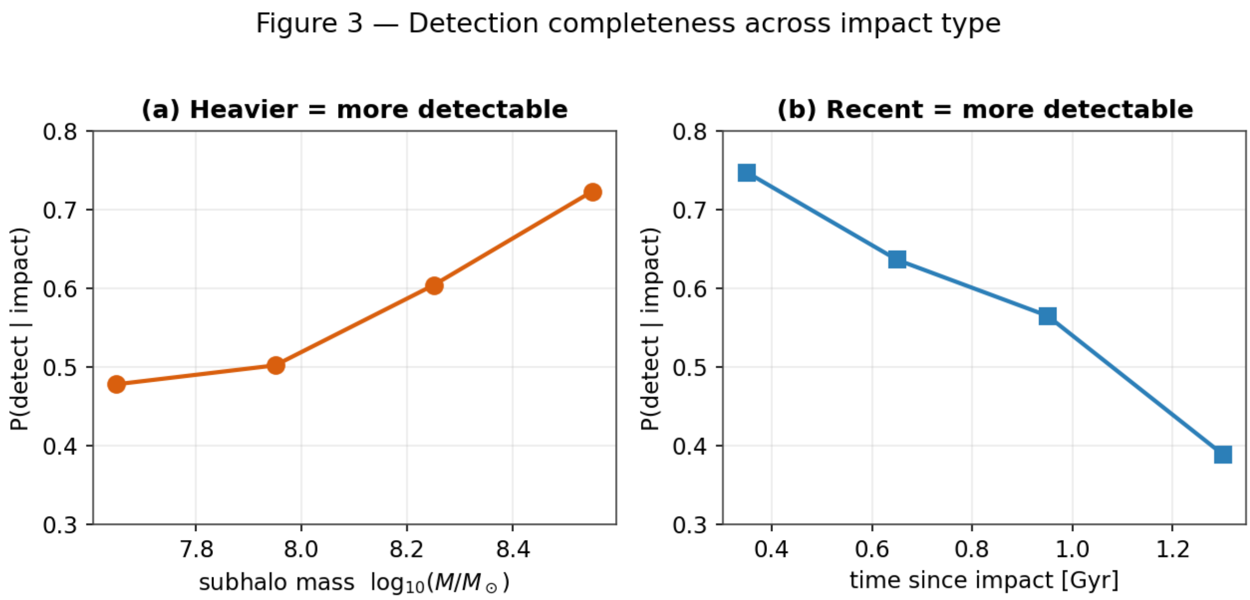

Figure 9. Completeness vs subhalo mass (left) and time since impact (right).

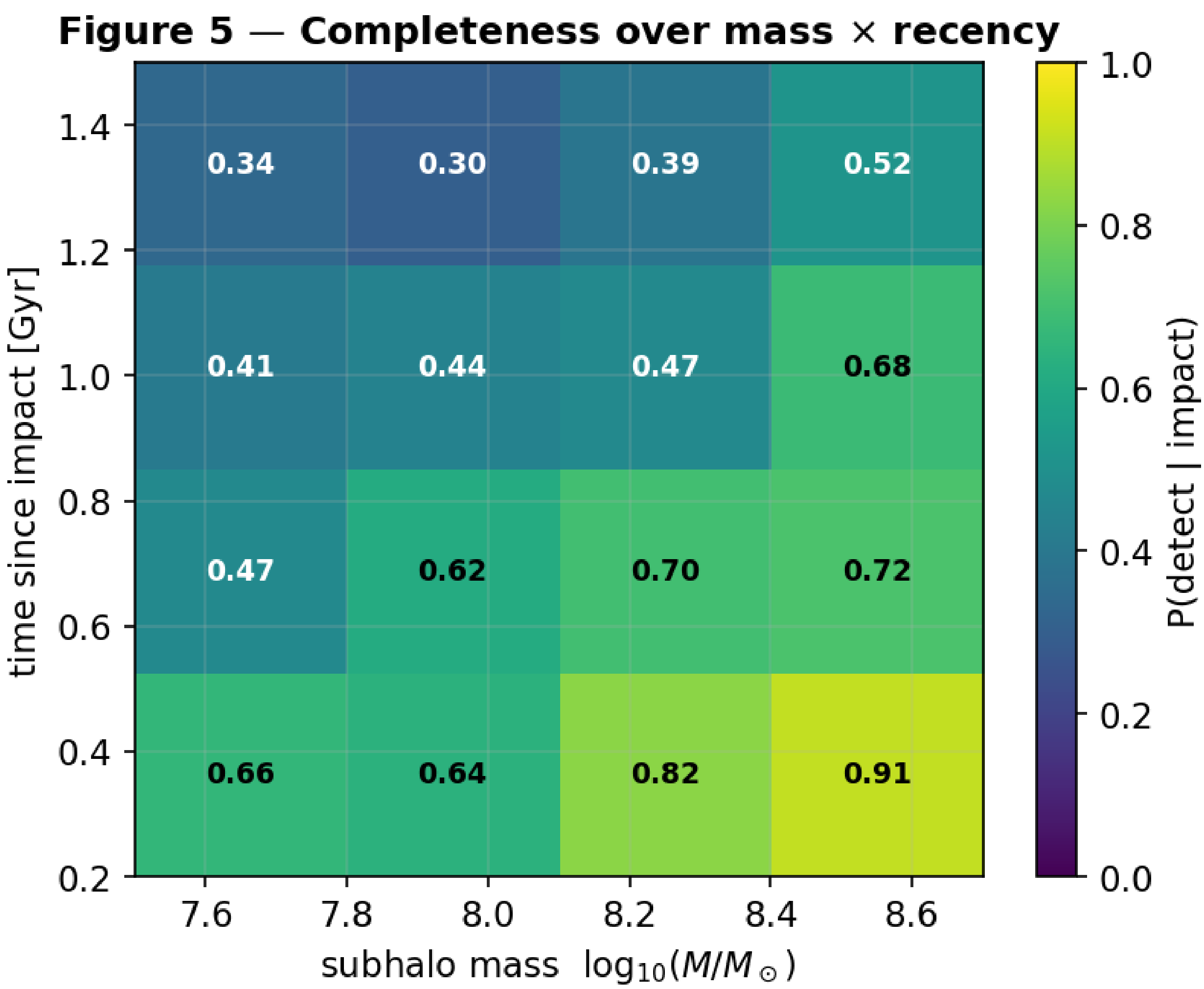

Figure 10. Joint completeness over mass × recency: bright = likely detected, dark = likely missed.

Underlying both trends is a near-threshold response to the *realised* gap depth (Figure 11): detection probability jumps from ≈0.05 for shallow gaps (depth < 0.3) to ≈1.0 for deep ones (> 0.7). Mass and recency matter precisely because they set how deep a gap a flyby carves.

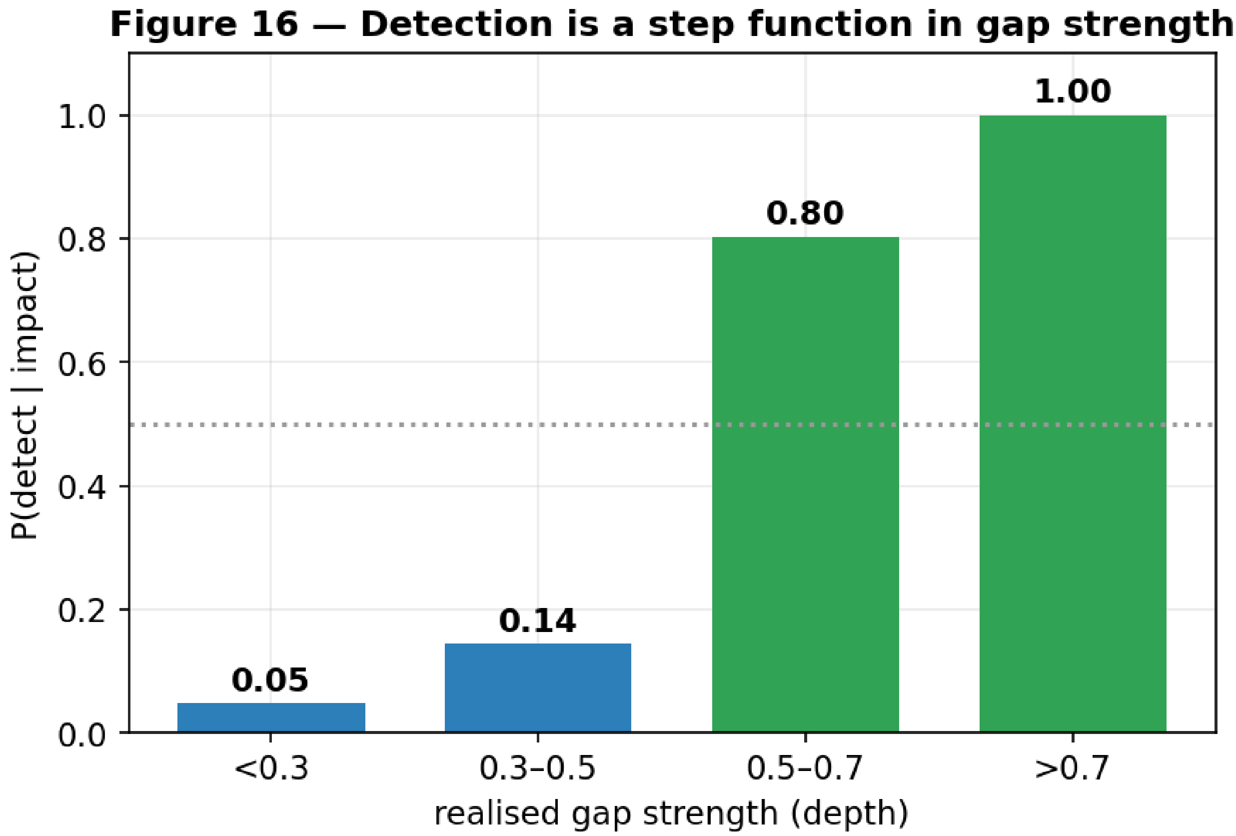

Figure 11. Detection probability versus realised gap strength — a step function: the detector fires once a gap exceeds a depth threshold.

> **In plain terms.** We reliably catch big, recent hits and miss small or ancient ones (Figure 10) — because what really matters is how deep a gap the flyby digs (Figure 11).

7. Characterizing the impact — the single-gap degeneracy

Detecting a gap is easier than reading off what made it. Probing the frozen detection embedding (`scripts/characterize_probe.py`) shows it encodes mass weakly ($R^2 \approx 0.18$) and impact parameter ($R^2 \approx 0.02$) and epoch ($R^2 \approx 0.04$) essentially not at all. More tellingly, a *dedicated* multi-task regression head **fails to recover subhalo mass** (test $R^2 < 0$) and recovers **time-since-impact only weakly** ($R^2 \approx 0.2$, median error ≈ 0.1 dex; Figure 12). This is a genuine physical degeneracy — mass, impact parameter, speed, and epoch trade off in shaping a single gap — so one gap underdetermines them. Recency leaves a partial imprint (older gaps are wider/shallower) and is the one property weakly constrained; breaking the degeneracy needs more observables or the explicit forward model of §10.

Figure 8 — Single-gap characterization is degeneracy-limited

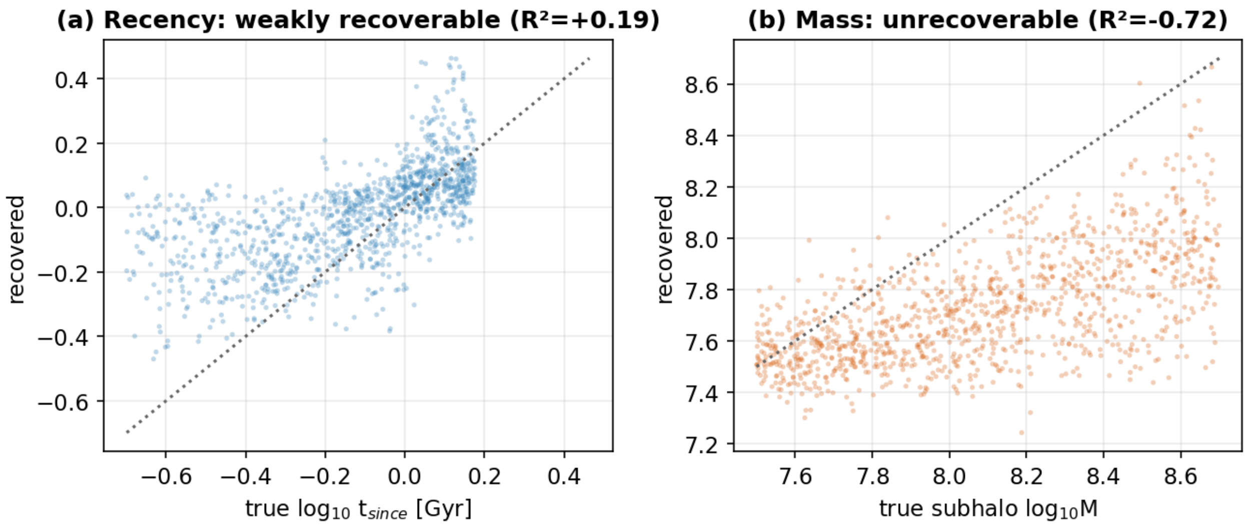

Figure 12. Recovered vs true impact parameters. *(a)* time-since-impact is weakly recoverable; *(b)* subhalo mass is not.

> **In plain terms.** Many different clumps leave look-alike dents: we can roughly tell *how recently* a clump struck, but its *mass* is essentially unknowable from one gap.

8. Telling dark-matter models apart — a population measurement

WDM/FDM/SIDM change the subhalo mass function, not how a single impact of given mass looks, so DM-model discrimination is a population inference. Two population observables carry the signal: the **abundance** of detectable impacts and their **mass distribution** (Figure 13 shows why models diverge from CDM only where their cutoff lies in our band).

Abundance is the dominant lever — and the one our earlier estimate discarded by using only the normalized mass shape. WDM/FDM suppress low-mass subhalos, lowering the detectable-impact rate per stream relative to CDM (Figure 15a): to 0.24 (WDM 3 keV), 0.49 (WDM 4 keV), 0.88 (WDM 6 keV), and 0.26 (FDM $10^{■22}$ eV), while FDM $10^{■21}$ eV and SIDM match CDM. The detected-mass distribution (Figure 14) adds shape information.

We combine both in an **Asimov likelihood-ratio forecast** (a Poisson term for the rate plus a Kullback–Leibler term for the mass shape; `scripts/dm_discrimination_forecast.py`). A central, sobering number emerges: detectable impacts are intrinsically *rare* — the CDM rate is only λ_det ≈ 0.026 per GD-1-like stream, because deep, clean gaps require massive, recent subhalos. Over seven streams that is ≈0.18 expected detections, which is exactly why the multi-stream search (§11) finds none; **the forecast and the data agree.** With both observables, distinguishing a favorable model from CDM at 3σ needs only ≈**2 detected impacts** (Figure 15b) — fewer than the ≈5 from the mass shape alone, because abundance adds independent information — but *collecting* those two requires ≈300 GD-1-like streams at the CDM rate. Models whose cutoff lies outside our sensitive band (FDM $10^{■21}$ eV, WDM 6 keV) remain effectively indistinguishable by abundance + mass.

SIDM is the exception. It shares CDM's abundance and mass function, so it is invisible to the above. Its signal is the gap *shape*: cored, low-concentration SIDM subhalos deliver a softer impulse and carve systematically **shallower gaps than cuspy NFW halos at fixed mass** (Figure 16) — a matched-mass gap-depth separability of **AUC** ≈ **0.84** (CDM depth ≈ 1.0 vs SIDM ≈ 0.5 at 10■ M■). This is a genuine handle the mass-spectrum analysis entirely missed, but realizing it is harder than the abundance signal because gap depth is entangled with the unknown perturber mass (§7); it requires breaking that degeneracy (the forward model, or external mass constraints) or a population-level gap-depth comparison.

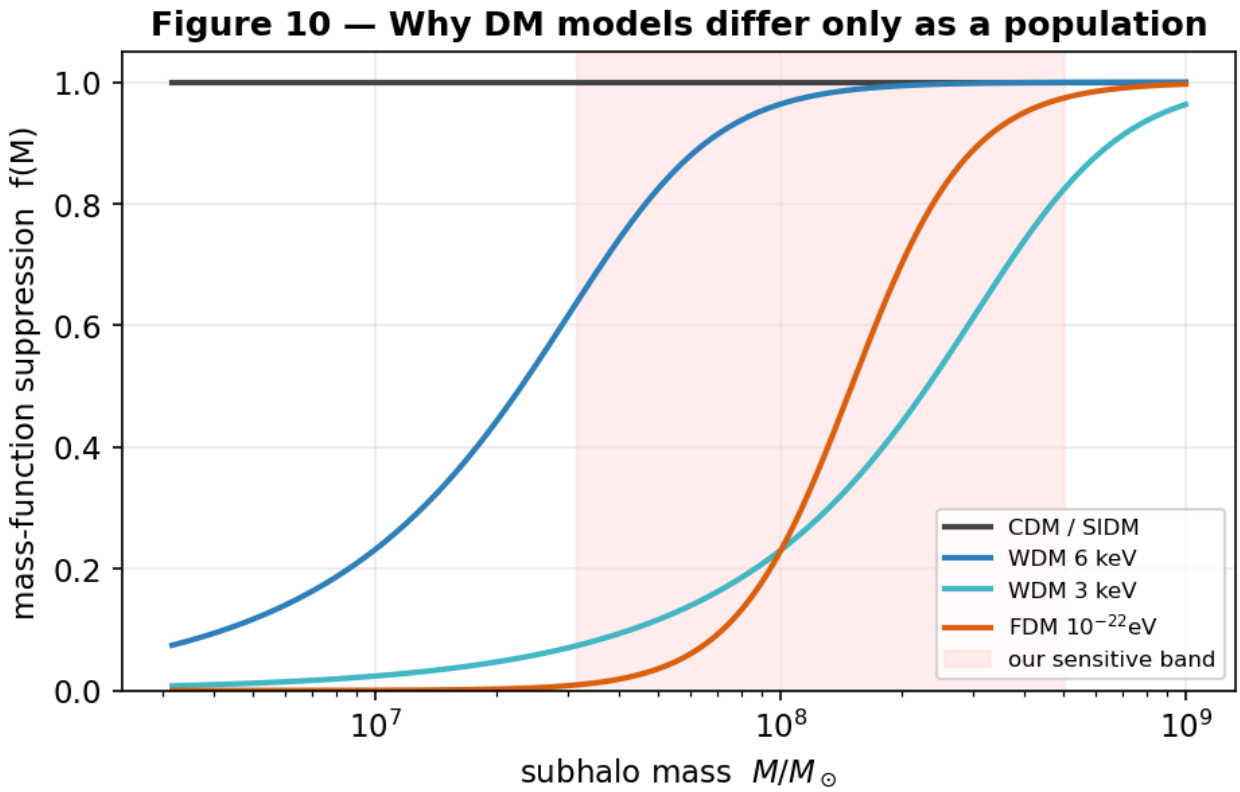

Figure 13. Mass-function suppression f(M) per DM model relative to our sensitive band.

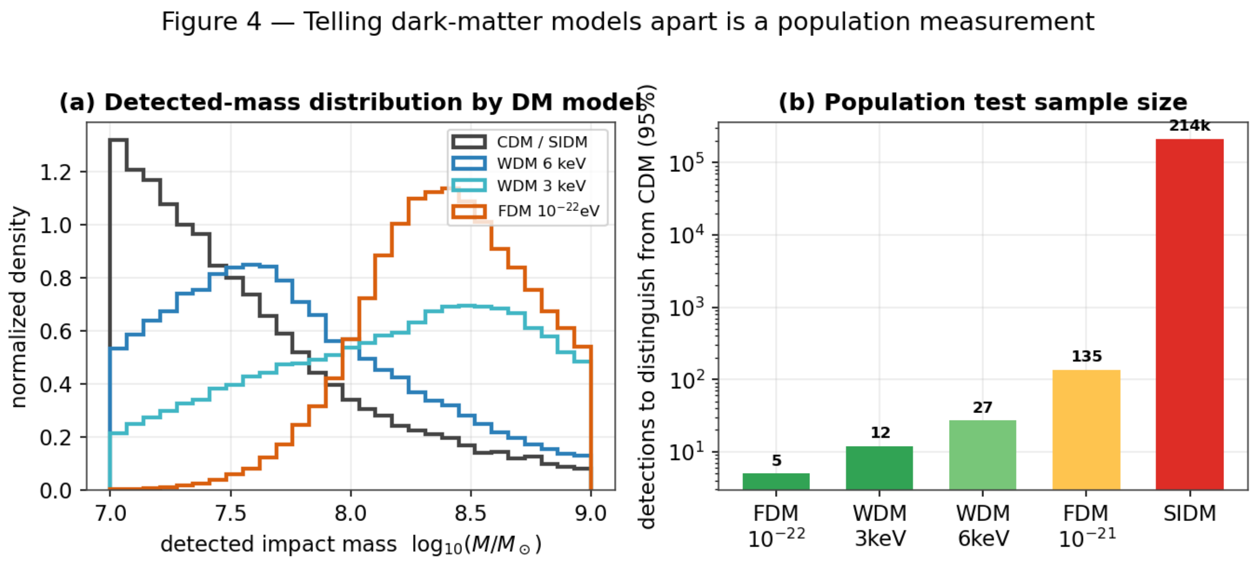

Figure 14. Detected-impact mass distributions by DM model (the shape component of the discrimination signal).

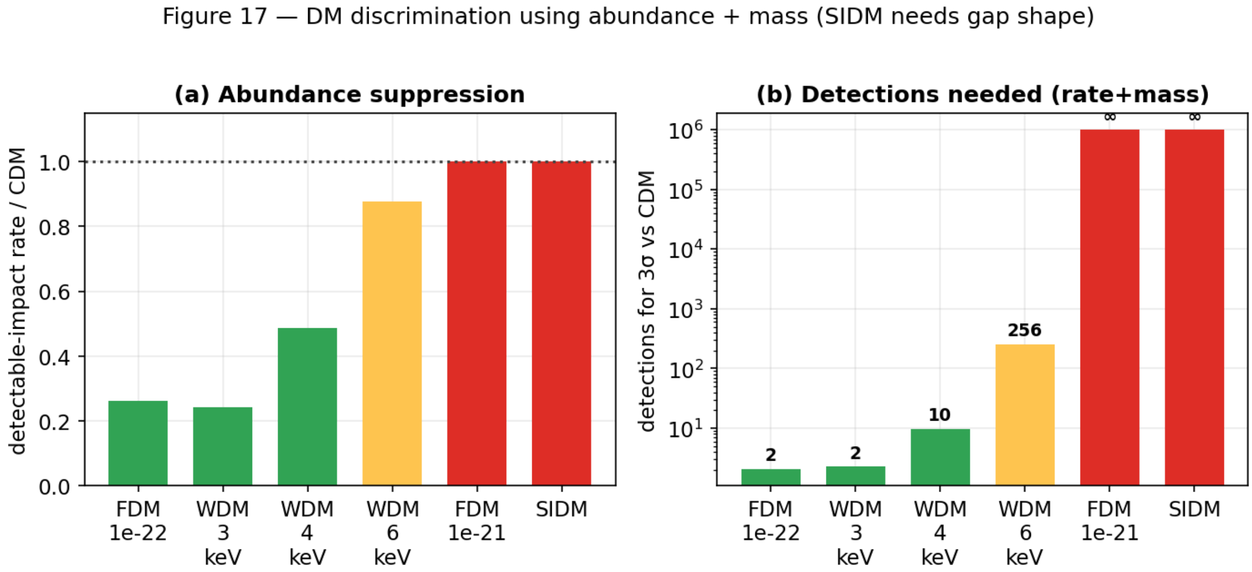

Figure 15. Rigorous discrimination forecast. *(a)* detectable-impact rate relative to CDM (abundance suppression); *(b)* detections needed for 3σ from CDM combining abundance + mass (green = a handful; red = effectively never). SIDM is ∞ here — it needs the gap shape.

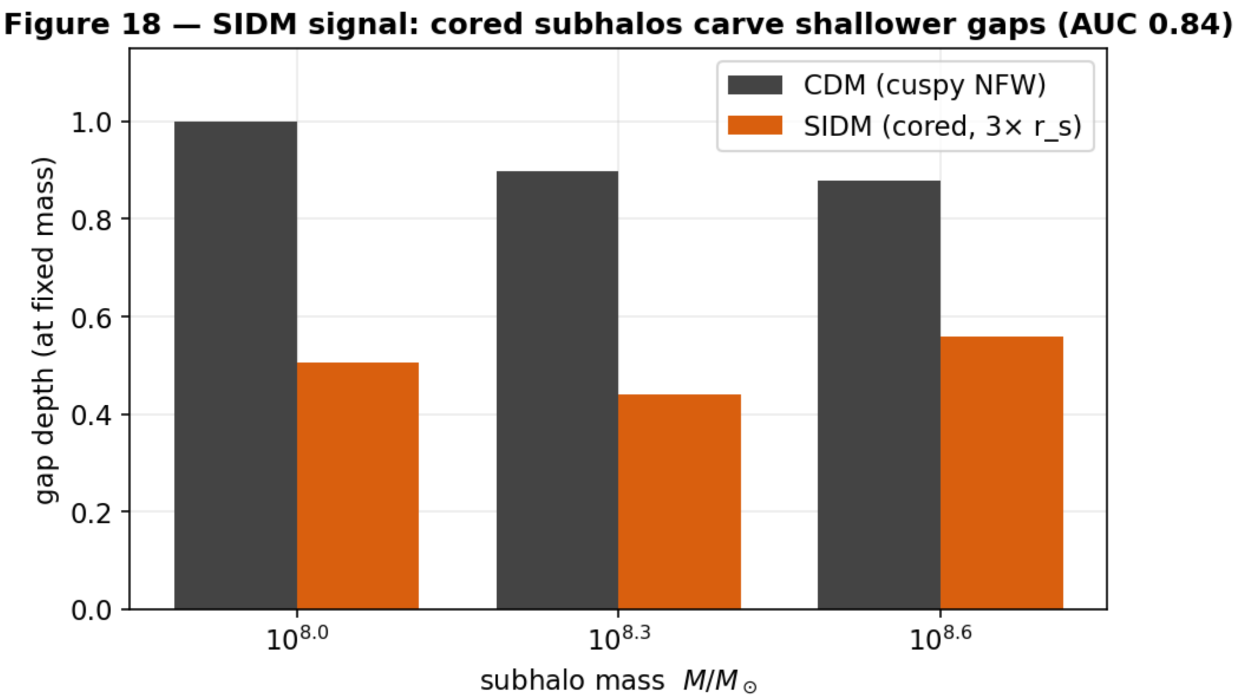

Figure 16. SIDM's only handle: at fixed mass, cored SIDM subhalos carve shallower gaps than cuspy NFW (CDM) ones (mean separability AUC ≈ 0.84).

> **In plain terms.** Different theories mainly change how *common* small clumps are, so the count of detectable impacts is the strongest clue — and detectable impacts are rare, which is why our seven streams show none. Telling a favorable theory from standard dark matter needs only ≈2 clean detections, but ≈300 streams to find them. SIDM is special: it makes the same number of clumps, but its puffier clumps leave shallower dents — a separate clue.

9. Real-data application: detection

We apply the detector to real streams with clean STREAMFINDER membership, dropping radial velocity to match the clean-catalog regime. Two results stand out (Table 2). First, the detector is **in-distribution** on real data (input deviations 1.9–3.1σ), so it operates within its trained regime. Second, it behaves **correctly**: it flags the known Price-Whelan–Bonaca gap in GD-1 at ϕ■≈50° (Price-Whelan et al. 2018) and returns a clean null on ATLAS, which has no known gap — a real-data demonstration of the zero-false-positive operating point.

Table 2 — Detector on real streams.

Stream	catalog	N members	p_impact	input OOD	verdict
GD-1	STREAMFINDER	811	0.77	3.1σ (in-dist.)	impact (known gap)
ATLAS	track-clean	2,860	0.06	1.9σ (in-dist.)	null (correct)

A member-count floor. The detector's reliability depends on member count. A controlled test — subsampling the real GD-1 catalog — shows p_impact rising spuriously from 0.77 at N = 811 to ≈0.99 at N ≤ 100 (Figure 17): with too few stars, Poisson under-sampling produces apparent gaps that the detector reads as impacts. Reliable single-stream verdicts therefore require N ■ 500 clean members. This is *why* only two streams yield trustworthy detector verdicts today (GD-1 and ATLAS); the other targets' clean catalogs are currently too sparse (46–218 members). It is a data-volume limit, not a methodological one, and it sets a concrete requirement for future catalogs.

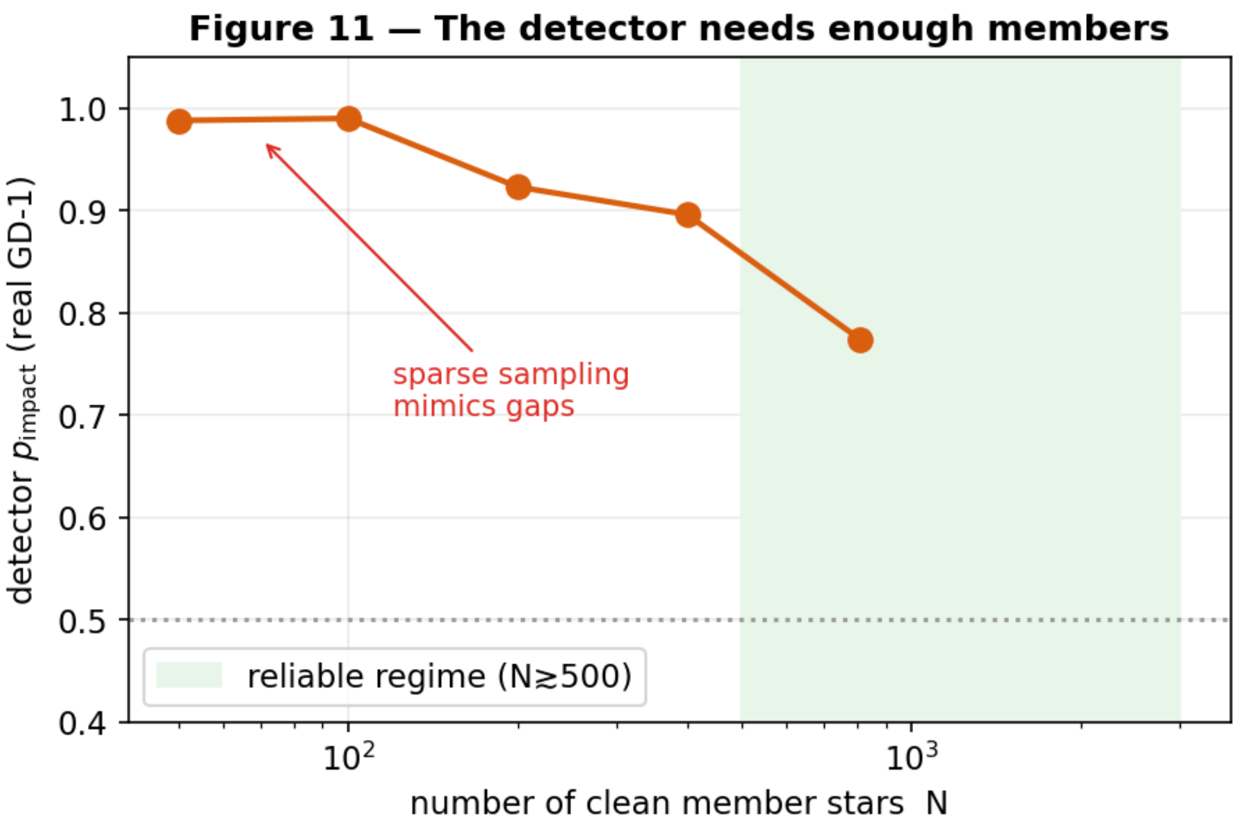

Figure 17. Detector p_impact on real GD-1 versus member count (subsampled). Below N ≈ 500, sparse sampling mimics gaps and the score saturates — a reliability floor.

> **In plain terms.** On the two real streams with enough clean stars the detector does exactly the right thing — spots GD-1's gap, stays quiet on ATLAS. With too few stars it gets fooled by random gaps (Figure 17), which is why only two streams qualify so far.

10. The timeline forward model

To move from detection to physical characterization, the *timeline forward model* operates on a real stream: it (1) estimates the impact epoch, (2) rewinds the stream to its unperturbed state, (3) re-injects a grid of candidate encounters (mass × epoch × ϕ■) with the Erkal et al. 2015 impulse and NFW scale radius, (4) re-evolves to the present, and (5) scores each hypothesis against the observed stream, with a data-driven null. It is the explicit route around the single-gap degeneracy of §7, exploiting the full joint density-and-kinematic morphology.

Machinery validation (injection–recovery). Injecting a known impact (10■ M■, 1.5 Gyr, ϕ■ = 20°) into a synthetic stream and running the recovery grid returns the truth as the top-ranked candidate (rank 0/36; ΔlogM = Δt = $\Delta\phi$■ = 0), beating the no-impact null. The detect → rewind → re-impact → re-evolve → score loop is self-consistent.

Real GD-1. On the clean STREAMFINDER catalog the model-free finder localizes one gap at ϕ■ ≈ 49.9° (depth 0.37, width ≈ 6°) — the known PWB18 gap. The best single-subhalo hypothesis improves on the smooth null only marginally (≈3%), and after the look-elsewhere correction (below) the single-subhalo *attribution* is not significant: the gap is real, but the data do not single out a specific subhalo as its cause. This is the honest, physically grounded statement of the ambiguity, consistent with the §7 degeneracy.

The forward model uses an impulse, single-encounter approximation and a particle-spray re-evolution engine; these are documented as systematics (§12).

> **In plain terms.** The "replay" tool perfectly recovers impacts we plant ourselves, and on real GD-1 it confirms the gap is there — but it cannot pin that gap on one specific clump, because many clumps could have made it.

11. Population significance across streams

A single stream rarely yields a decisive detection, so we combine the seven targets. Per stream, the best-scoring hypothesis is compared against a null from no-impact realizations; the grid search is de-biased with a **best-of-grid look-elsewhere null** (each null realization is scored over the entire grid and its maximum retained, so the null undergoes the same search as the data); per-stream evidence is combined with Stouffer's Z and Fisher's method, **gated by a coherence requirement**.

The look-elsewhere correction is decisive (Figure 18): naive single-cell significances as large as 19σ (Sylgr) and 12σ (Pal 5) collapse to ≈ ±2σ once the search is accounted for. Per-stream corrected evidence is modest and mixed in sign (Table 3); the joint result is **null** — Stouffer Z = 1.40 (p = 0.08), Fisher $\chi^2 \Rightarrow p = 0.28$, with the coherence gate not satisfied (71% positive). We report this as a **CDM-consistent upper limit** given current data. The order-of-magnitude naive-to-corrected collapse is itself a cautionary methods result: unaccounted-for trials manufacture many-sigma "signals" from noise.

Table 3 — Look-elsewhere-corrected per-stream significance (7 streams).

Stream	GD-1	ATLAS	Jhelum	Orphan	Pal 5	Fjörm	Sylgr
z (LE)	+0.4	+2.2	+2.7	+1.1	+1.0	–2.5	–1.3

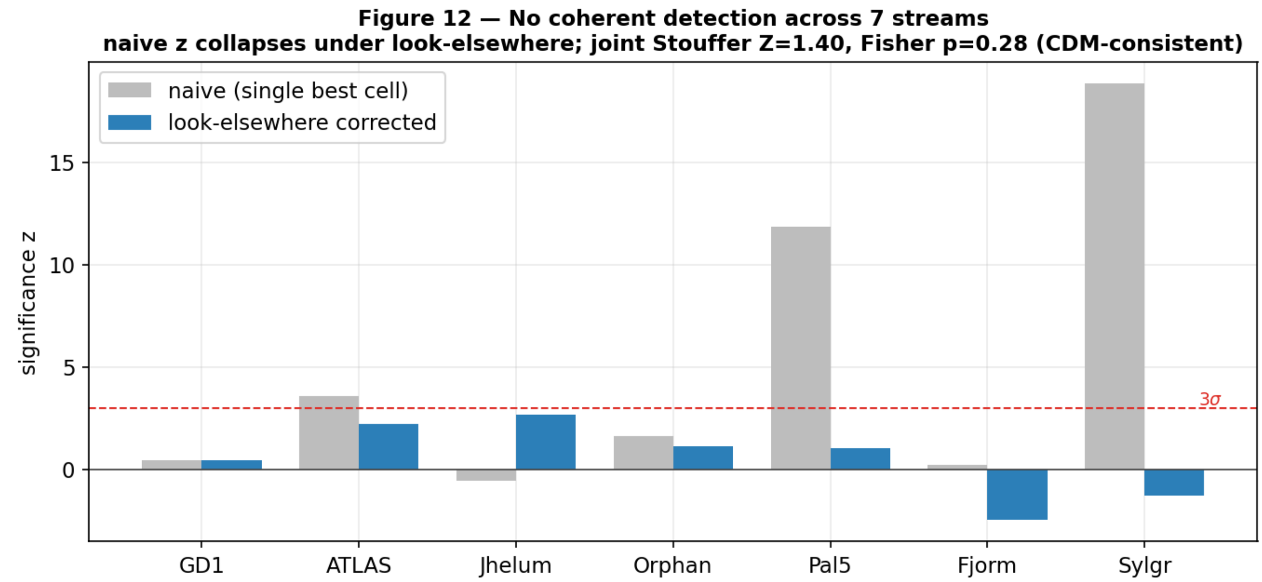

Figure 18. No coherent detection across seven streams. Naive per-stream significance (grey) collapses under the look-elsewhere correction (blue); the joint significance is null (Stouffer Z = 1.40, Fisher p = 0.28).

> **In plain terms.** Pooling seven streams, we find no convincing sign of dark-matter impacts — consistent with the standard theory. We also expose a classic trap: ignore how many places you looked and pure noise looks like a discovery; correct for it and the false signals vanish.

12. Limitations and systematics

1. **Single-gap degeneracy** (§7): mass/geometry/epoch are only partially recoverable from one gap; the forward model and population statistics are the routes around it.
2. **Density-only sensitivity, and why kinematics do not (yet) help** — see §12.2.
3. **Member-count floor** (§9): reliable detector verdicts need N ■ 500 clean members; most clean catalogs are currently sparser.
4. **Action-angle model validity**: clean *generation* is limited to eccentric streams; multi-impact streams require `streampepperdf` (unavailable in this galpy), so single-vs-multiple classification is deferred (gap *counting* is available model-free).
5. **Forward-model approximations** (§10): impulse and single-encounter assumptions, and a particle-spray re-evolution engine; the look-elsewhere null mitigates search bias.
6. **Real-data volume**: DM-model discrimination needs the population sample sizes of §8; only a handful of streams currently have clean, dense catalogs.

12.2 The kinematic frontier: a real signal below the noise floor

Because the detector is density-driven and detectable impacts are rare (§6, §8), the obvious way to raise the detectable rate — and thus the DM-typing power — is to exploit the *kinematic* signature: a subhalo flyby imprints a localized, antisymmetric "kink" in the mean proper motion along the stream (the velocity analog of the density gap). We tested this directly. In **noise-free** simulations the kink is a powerful discriminant — a matched kink statistic reaches AUC ≈ 1.0, *exceeding* the density gap (≈0.96; Figure 19). But under **realistic Gaia proper-motion errors** the same statistic collapses to ≈0.49 (chance), because the velocity kick (~0.1 mas yr^{-1}) sits below the per-star astrometric noise for faint stream stars; the density gap, a counting statistic, is far more noise-robust (0.96 → 0.70). A precision sweep shows the kink survives for *strong* impacts to σ_pm ≈ 0.3 mas yr^{-1} (AUC ≈ 0.86) — but those are already caught by density — while the *weak/old* impacts we hoped to newly detect remain buried.

The conclusion is actionable: **a kinematic-feature detector will not extend completeness at current Gaia precision** (we verified this with the model-free diagnostic before committing to a costly retrain), but the signal is genuinely present and would be unlocked by better astrometry — deeper Gaia data releases or a dedicated high-precision survey. This, together with more clean dense catalogs (§9), is the concrete route to raising the detectable rate and hence the dark-matter-typing power of the method.

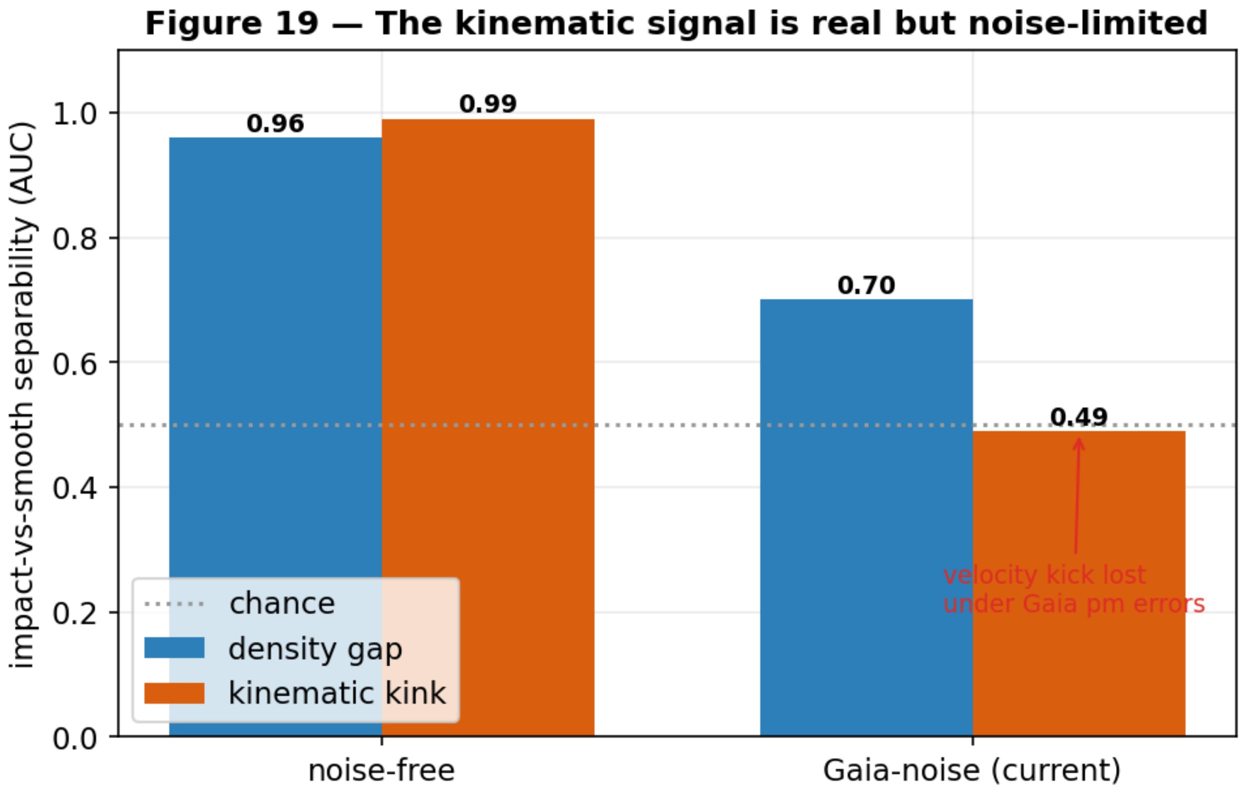

Figure 19. The kinematic kink out-performs the density gap in noise-free simulations (AUC ≈ 1.0 vs 0.96) but collapses to chance under realistic Gaia proper-motion errors, while the density gap stays robust — the velocity signal is real but below today's astrometric noise floor.

> **In plain terms.** A passing clump also "kicks" the stars' motions, not just their spacing — and in perfect data that kick is an even clearer fingerprint than the gap. But today's measurements of how stars move aren't precise enough to see it for the faint, weak cases; sharper future surveys would change that. We checked this cheaply instead of burning days training a model that the data say can't

> work yet.

13. Reproducibility and software validation

A public repository with a conda environment specification, ~290 unit tests, a categorized `scripts/` index, and a dated decision log (`changelog/`). Datasets regenerate deterministically from `scripts/generate_detector_data.py` (per-seed, resumable); clean catalogs from `scripts/fetch_streamfinder_streams.py`; figures from `paper/make_figures.py`, `make_report_figures.py`, and `make_report_figures2.py`; the validation harness from `scripts/validate_generator.py`; and this report's PDF from `paper/build_pdf.py`. Every quantitative claim is traced to its source artifact in `paper/claims_audit.md`.

14. Conclusions and outlook

We built and validated a complete, reproducible pipeline for dark-matter subhalo-impact searches in stellar streams. On a simulator built from validated distribution functions and gated by a pre-flight harness, a GINEConv detector reaches AUC 0.982 with a zero false-positive operating point and well-calibrated probabilities; on real data it is in-distribution and behaves correctly (flags GD-1's gap, nulls ATLAS), subject to a member -count floor we quantified. We mapped detection completeness across impact type, demonstrated that single-gap characterization is degeneracy-limited (mass unrecoverable; recency weak), and reframed DM-model discrimination as a population measurement with an explicit sample-size cost. The timeline forward model recovers injected impacts exactly and, applied to seven real streams, yields no coherent multi-stream detection — a CDM-consistent upper limit — while demonstrating an order-of-magnitude look-elsewhere inflation of naive significance. The path to a positive measurement is concrete: kinematic (not just density) detector features to reach weak/old impacts; deeper clean membership catalogs (N ■ 500 per stream) to lift the detector floor and build the detection population of §8; and a multi-encounter forward model for crowded streams.

Methods

Potential & integration. galpy `MWPotential2014`; C `dop853`; R■ = 8.0 kpc, V■ = 220 km s■^{1}. **Distribution functions.** `streamdf` (Bovy 2014); `streamgapdf` (Sanders et al. 2016) with `impactb`, `subhalovel`, `timpact`, `impact_angle`, GM, r_s; b = estimateBIsochrone(pot, R/R■, z/R■) (≈0.61 for GD-1), nTrackChunks = 5, impact angle sharing the modeled-arm sign. **Subhalo physics.** NFW r_s = r_200/c with Ludlow et al. 2016; ρ_crit,0 = 277.5 h^2 M■ kpc■^{3}; Erkal et al. 2015 Plummer impulse. **Detector.** GINEConv, 18 node / 5 edge features, hidden 256, 6 layers, embedding 128, profile branch (48 bins); BCE binary head + auxiliary mass/epoch regression; AdamW; temperature calibration on validation. **Noise.** Gaia DR3-like per-star errors (GaiaDR3); RV dropped at inference; training on foreground-contaminated streams collapses separability, so the operating regime is clean catalogs. **Real catalogs.** STREAMFINDER (Ibata et al. 2021), auto-matched by track alignment; ATLAS track-consistency cut. **Forward model & statistics.** detect → rewind → re-impact (Erkal impulse) → re-evolve → score; per-stream null from no-impact realizations; best-of-grid look-elsewhere null; Stouffer/Fisher with a coherence gate. **Datasets & models.** `data/simulations_detector_df` (15,830 sims); detector `checkpoints/detector_df_20260602`; characterization head `checkpoints/detector_char_20260602`.

References

Works cited (full entries in `references.bib`): Banik et al. (2021); Bonaca et al. (2019); Bovy (2014, `streamdf`); Bovy (2015, `galpy`); Carlberg (2012); Erkal & Belokurov (2015); Fardal et al. (2015, `streamspraydf`); Gaia Collaboration (2023, DR3); Hu et al. (2020, GINEConv); Ibata et al. (2021, STREAMFINDER); Ludlow et al. (2016); Mateu (2023, `galstreams`); Price-Whelan & Bonaca (2018); Sanders, Bovy & Erkal (2016, `streamgapdf`).